FAR SIDE *southern hemisphere*

Mare Orientale

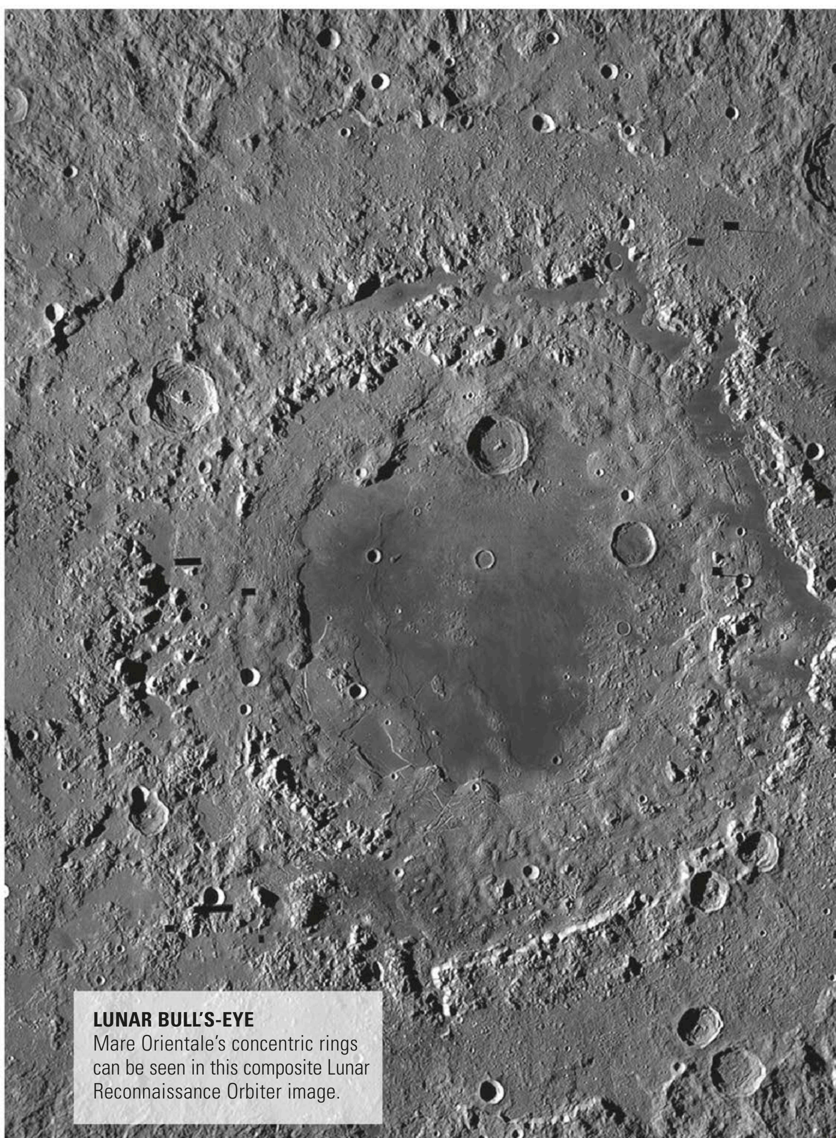
TYPE Multiring basin

AGE 3.8 billion years

DIAMETER 560 miles (900 km)

This multiring basin is half the size of the near side's Mare Imbrium. It lies on the eastern limb of the far side, and from Earth, the Montes Rook—the innermost eastern portion of the three distinct rings—can be clearly seen. This giant lunar bull's-eye was formed by a massive asteroid, and two theories have been proposed to explain the rings. The first has the impact excavating a deep transient crater. The cracked inner walls of this crater would have been unable to support the weight of surrounding crust, so the rock slumped into the hole, guided by a series of concentric fault systems that account for the rings that remain. Not only was most of the crater filled in, but the break-up of subsurface rock allowed lava from far below the lunar surface to seep up and fill in the central regions. However, the highland crust is about 37 miles (60 km) thick, and rock from below that depth should have been excavated, but this deep rock has not been found. Alternatively, the seismic shocks generated by the massive impact could have briefly turned the surrounding rocks into a fluidized powder. Tsunami-type waves moved out through the pulverized rock but quickly became frozen, resulting in three clearly visible mountain rings.

ROOK AND CORDILLERA MOUNTAINS
Orientale is surrounded by two huge circular mountain ranges. The outer range is called Montes Cordillera (above right) and the inner one is called Montes Rook (lower left).



LUNAR BULL'S-EYE
Mare Orientale's concentric rings can be seen in this composite Lunar Reconnaissance Orbiter image.

FAR SIDE *southern hemisphere*

South Pole–Aitken Basin

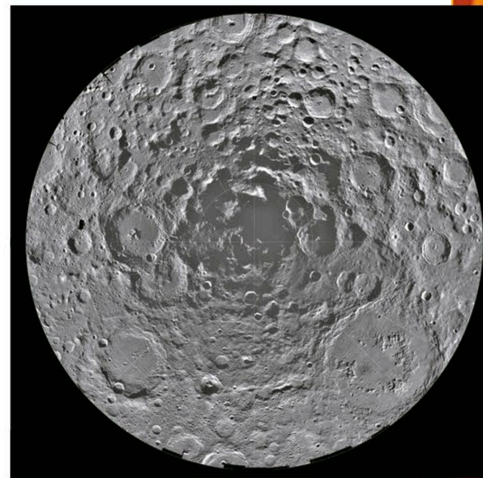
TYPE Impact crater

AGE 3.9 billion years

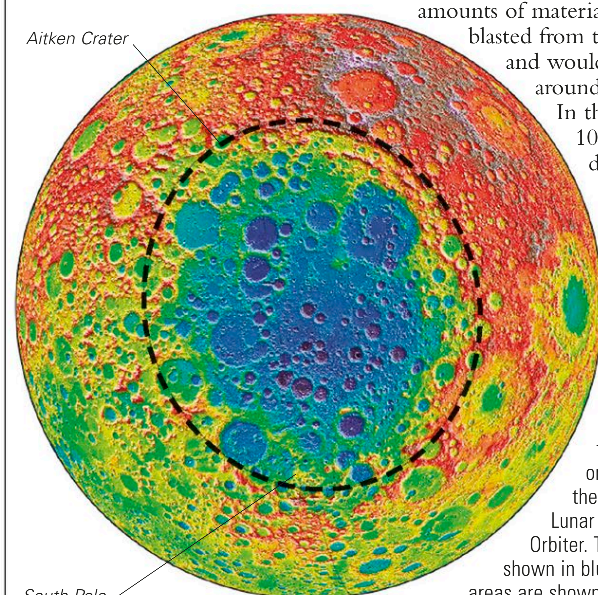
DIAMETER 1,550 miles (2,500 km)

The South Pole–Aitken Basin is an immense impact crater lying almost entirely on the far side of the Moon. It stretches from just above the South Pole to beyond the Aitken Crater, which is close to the center of the far side. South Pole–Aitken is a staggering 1,550 miles (2,500 km) in diameter and is over 7.4 miles (12 km) deep. It is one of the largest craters in the solar system and is comparable in size to the largest craters on Mars. Its diameter is about 70 percent of that of the Moon itself. The asteroid that produced it would have been over 60 miles (100 km) across.

Even though the basin was first discovered in 1962, detailed investigation only started when the Galileo spacecraft imaged the Moon in 1992, while on its way to Jupiter. The South Pole–Aitken Basin looked darker than the rest of the far-side highland rocks, indicating that the lower-crustal rocks at the bottom of the deep crater were richer in iron than normal lunar surface material. Iron oxide and titanium oxide abound. Impact geophysicists are convinced that a normal impact could not have produced a crater this large without digging up large amounts of rock from the mantle that lies below the lunar crust. It may be that the crater was produced by a low-velocity collision, with the impactor coming into the surface at a low angle. Huge amounts of material would have been blasted from the lunar surface and would have moved off around the Moon's orbit. In the subsequent 10 million years, this debris would have collided with the Moon, producing many new craters.



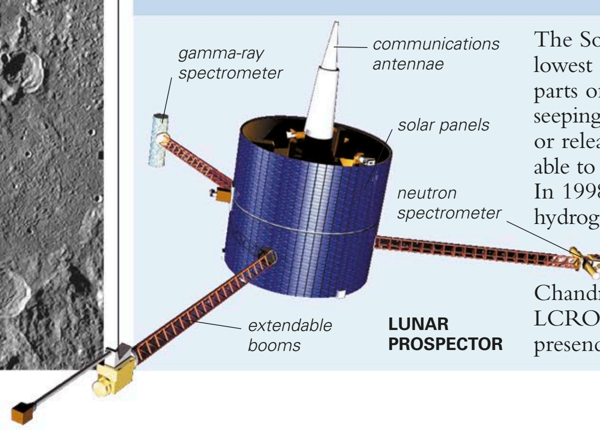
SOUTH POLE
The massively cratered, cold lunar South Pole can only be glimpsed tangentially from Earth. NASA's Clementine mission provided the first detailed map of the region in 1994.



LARGEST KNOWN IMPACT CRATER
The South Pole–Aitken Basin, the largest impact scar on the Moon, is outlined on this relief map from the altimeter on NASA's Lunar Reconnaissance Orbiter. The lowest areas are shown in blue, while the highest areas are shown in red and brown.

EXPLORING SPACE

LOOKING FOR WATER



LUNAR PROSPECTOR

The South Pole-Aitken is one of the lowest regions on the Moon, and parts of it never see the Sun. Water seeping up from cracks in the mantle or released by an impact will not be able to escape from these "cold traps." In 1998, Lunar Prospector found hydrogen, thought to be from the breakup of water ice, within these traps. Both the Chandrayaan-1 mission (2008) and LCROSS (2009) also indicate the presence of water in this region.